LEARN CODING

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github.com/ale66/learn-coding

AUXILIARY VARIABLES

TOPICS

- counters
- accumulators
- trailers

COUNTERS

A variable, normally of type integer, that is incremented as we *parse* an iterable

Example: compute list size/length

```
1  l = 0
2
3  for f in fruits:
4
5    l = l +1
6
7  print(f'The length of list {fruits} is {l}')
```

Syntax: `l += 1` is shorthand for `l = l + 1`

ANONYMOUS VARIABLES

In fact, we are not operating on `f`

it can be omitted in favour of an *anonymous variable* `_`

```
1 l = 0
2
3 for _ in fruits:
4
5   l += 1
6
7 print(f'The lenght of list {fruits} is {l}')
```

ACCUMULATORS

A variable to record quantities seen during the iteration

Example: the average of an arbitrary list of positive numbers

```
 1  l = 0
 2  accumulator = 0
 3
 4  for val in mylist:
 5    l += 1
 6    accumulator += val
 7
 8  if l > 0:
 9    average =  accumulator / l # this will be a float
10
11  print(f'There are {l} values and their average is {average}')
```

TRAILING VARIABLES

A variable that *remembers* past values during an iteration

Example: find the biggest *temperature increase* in a list

```
# start with the minimum possible value
max_day_on_day_change = 0

# initialise the trailer variable
trailer = temperatures_this_month[0]

for temp in temperatures_this_month:

  # do your thing here...

  # we are finished with this value, assign it to the trailer
  trailer = temp

print(f'The maximum day-on-day increase has been {maxincrease} degrees')
```

PROGRAMME COMPREHENSION

To grasp how trailers work, put extra `print()` commands to see what the variables contain

Rename variables:

```
1  yesterday = temperaturelist[0]
2
3  for today in temperaturelist:
4
5    current_increase = today - yesterday
6
7    ...
8
9    yesterday =  today
```

A MORE COMPLETE ENCODING

```
1 temperatures_this_month = [27, 21, 16, 21, 19, 18, 19, 12, 15, 12]
2
3 '''
4 Question:
5 which day of the month has seen
6 the steepest temperature change (either increasing or decreasing)?
7 '''
```

```python
# start with the minimum possible value
max_day_on_day_change = 0

# initialise the trailer variable
trailer = temperatures_this_month[0]

for temp in temperatures_this_month:

  current_change = temp - trailer

  if abs(current_change) > max_day_on_day_change:

    max_day_on_day_change = abs(current_change) # a new max is found

  # we are finished with this value, assign it to the trailer
  trailer = temp

print(f'The maximum day-on-day increase has been {max_day_on_day_change} degrees')
```

Run it on Pythontutor

QUIZ

Complete code for searching a given number in a sequence

```
1  KEY = 22
2
3  mylist = [1, 6, 3, 9, 10]
4
5  ___ item ___ mylist:
6    if ___ == ___:
7      found = True
8    else:
9      found = False
```